

Table of Contents

Section	Title	Page No.
Chapter 1 – Industry Review		
1.1	Current Practices	4
1.2	Background	4
1.3	Problem Statement	4
1.4	Impact on Business	4
1.5	Literature Review	5
Chapter 2 – Dataset and Domain		
2.1	Data Sources	6
2.2	Dataset Overview	6
2.3	Data Dictionary and Pre-Processing Data Analysis	7
2.4	Data Cleaning and Preparation	7
Chapter 3 – Exploratory Data Analysis (EDA)		
3.1	Distribution of District Delivery Rate	8
3.2	Office Type & Delivery Analysis	8
3.3	Infrastructure Distribution by State	9
3.4	Complaint Trend Analysis	9
3.5	Correlation Analysis	10
3.6	Bivariate Analysis – BO Ratio	10
3.7	Box Plot – Delivery Rate by Office Type	10
Chapter 4 – Feature Engineering		
4.1	Data Transformation & Encoding	11

4.2	Feature Scaling	11
4.3	VIF Analysis & Feature Selection	11
Chapter 5 – Modelling Approach and Assumptions		
5.1	Modelling Approach	12
5.2	Models Implemented	12
5.3	Model Results and Performance Evaluation	12
5.4	Model Comparison Summary	15
5.5	Linear Regression Assumptions Check	16
5.6	Feature Importance Analysis	16
5.7	Business Metrics & Success Criteria	17
5.8	Advanced Ensemble Models (XGBoost & CatBoost)	17
5.9	Comparative Feature Importance Insights	19
5.10	Model Interpretation & Business Insight Validation	19
Chapter 6 – Deployment: Streamlit-Based Decision Support System		18
6.1	Application Overview	20
6.2	Key Features	20
6.3	System Workflow	24
6.4	Business Impact	24
6.5	Key Advantages	24
Chapter 7 – Conclusion & Scope for Enhancement		25

Chapter 1 – Industry Review

1.1 Current Practices

The postal and logistics sector plays a vital role in connecting urban and rural regions by delivering mail, parcels, and government services. India Post manages three main types of postal offices:

- Head Post Offices (HO) – Administrative and operational hubs handling high-volume mail processing
- Sub Post Offices (PO) – District-level service centers for general postal and financial services
- Branch Post Offices (BO) – Rural last-mile delivery points reaching remote communities

With growing operational data, postal organizations are now adopting data analytics and machine learning to evaluate infrastructure distribution and improve delivery efficiency.

1.2 Background

India Post operates one of the world's largest postal networks, with over 165,000 post offices serving both urban and rural populations across multiple administrative levels — circles, regions, divisions, and districts.

- Despite the large network, the link between infrastructure distribution and delivery performance is not well understood
- Some districts have many offices but still face operational inefficiencies
- Other districts achieve high delivery rates with relatively fewer offices

This project uses data-driven methods to analyze how infrastructure composition influences delivery efficiency at the district level.

1.3 Problem Statement

The current distribution of postal infrastructure across districts does not always align with delivery performance. Key issues include:

- Some regions have excess infrastructure, leading to underutilization
- Other regions have insufficient coverage, causing delivery delays and poor service
- There is no systematic analytical model to evaluate infrastructure efficiency at the district level

This project fills this gap by developing a data-driven infrastructure efficiency model that identifies patterns linking postal infrastructure composition to district-level delivery performance.

1.4 Impact on Business

Efficient infrastructure distribution directly impacts operational efficiency, service accessibility, and customer satisfaction. The model helps India Post to:

- Identify districts with insufficient postal coverage and prioritize expansion
- Detect underutilized offices and optimize resource allocation
- Improve delivery efficiency and reduce operational bottlenecks
- Support data-driven policy decisions for infrastructure planning
- Enable better last-mile delivery planning for the growing e-commerce sector

1.5 Literature Review

Paper 1 – Machine Learning for Postal Delivery Prediction

- Authors: Husejnović et al. (2025)
- Evaluates ML algorithms (Random Forest, SVM, XGBoost) for postal delivery performance prediction
- Relevance: Supports the use of ensemble learning methods for predicting postal delivery outcomes in this project

Paper 2 – CRISP-DM Data Mining Framework

- Authors: Wirth & Hipp (2000)
- Defines a six-stage structured framework: Business Understanding → Data Understanding → Data Preparation → Modelling → Evaluation → Deployment
- Relevance: This project follows the CRISP-DM methodology throughout its workflow

Paper 3 – Extremely Randomized Trees Algorithm

- Authors: Geurts et al. (2006)
- Introduces the Extra Trees algorithm — an ensemble that builds multiple randomized decision trees to improve accuracy and reduce variance
- Relevance: The Extra Trees Regressor achieved the highest prediction accuracy ($R^2 = 0.888$) in this project

Paper 4 – Data-Driven Optimization in Postal Logistics

- Source: USPS Office of Inspector General (2018)
- Discusses the role of predictive analytics, optimization algorithms, and ML in improving postal logistics and infrastructure planning
- Relevance: Highlights the strategic importance of data-driven approaches for postal infrastructure optimization

Chapter 2 – Dataset and Domain

2.1 Data Sources

- Three datasets were used in this project:
- Postal Infrastructure Dataset (Primary) – 1,65,627 records of individual post offices across India, sourced from publicly available India Post records. Includes office type, delivery status, district, state, and geo-coordinates.
- Statewise Dataset – 36 states/UTs with aggregate office counts (Head PO, Sub PO, Branch PO, Total PO) and computed ratios.
- Complaints Dataset – FY 2016–21 data on complaint volumes and complaint rates, used to contextualize service quality trends over time.

2.2 Dataset Overview

- A total of 163,014 records were retained after merging and cleaning. (from original 1,65,627)
- 749 unique districts across 31 states and union territories
- 3 office types: BO (Branch Post Office), PO (Sub Post Office), HO (Head Post Office)
- Target variable: `district_delivery_rate` — proportion of delivery-enabled offices per district (range: 0–1)
- Features include: administrative identifiers, infrastructure counts, office-type ratios, pincode zone, and state encoding

2.3 Data Dictionary and Pre-Processing

- The table below summarizes key variables in the analysis:

Column Name	Data Type	Description	Null %
circlename	object	23 unique postal circles	0%
regionname	object	50 unique regional offices	0.19%
divisionname	object	474 postal divisions	0%
officetype	object	3 types: HO, PO, BO	0%
delivery	object	Delivery / Non-Delivery (text)	0%
district	object	720+ districts	0.43%
statename	object	31 states and union territories	0.43%
latitude / longitude	float	Geo-coordinates of postal offices	7.25%
Head_PO / Sub_PO / Branch_PO	int	Office counts per state (from statewide data)	0%
bo_ratio / po_ratio / ho_ratio	float	District-level office type proportion	0%
district_delivery_rate	float	Target variable: delivery offices ÷ total offices	0%
pincode_zone / pincode_subzone	int	Zone classification based on pincode prefix	0%
state_encoded	int	Label-encoded state identifier (0–30)	0%

2.4 Data Cleaning and Preparation

- Missing values in latitude/longitude (~7.25%) were retained — these apply to remote Branch Offices and do not affect model features
- Rows with missing district or state names were dropped (~2,600 records removed to ensure data quality)
- The delivery column (Delivery/Non-Delivery text) was converted to binary: 1 = Delivery, 0 = Non-Delivery
- Data leakage prevention: 163,014 row-level records were collapsed into 749 district-level records before modelling
- Target variable computed as: $\text{district_delivery_rate} = \text{delivery_offices} \div \text{total_offices}$ per district
- Categorical variables (State, Office Type) were label-encoded for machine learning compatibility
- Infrastructure ratios (bo_ratio, po_ratio, ho_ratio) were derived to capture office composition per district

Chapter 3 – Exploratory Data Analysis (EDA)

EDA was conducted on both the row-level (163,014 records) and district-level (749 records) datasets to understand infrastructure distribution patterns, identify key correlations, and detect factors that influence delivery performance.

3.1 Distribution of District Delivery Rate

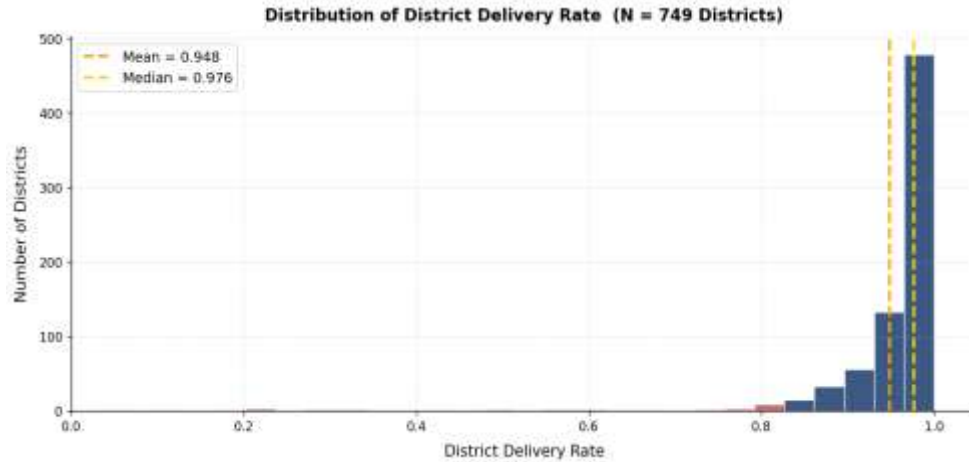


Fig 3.1 : Distribution of District Delivery Rate (749 Districts)

Key Insights:

- Distribution is heavily left-skewed (skewness ≈ -5.27) — most districts achieve delivery rates above 0.95
- Mean delivery rate = 0.948 ; Median = 0.976 — indicating strong nationwide delivery coverage
- Red bars (rate < 0.80) highlight under-performing districts — these are the primary targets for infrastructure optimization
- The ceiling effect and non-normal distribution justify the use of tree-based models over linear regression

3.2 Office Type Distribution & Delivery Analysis

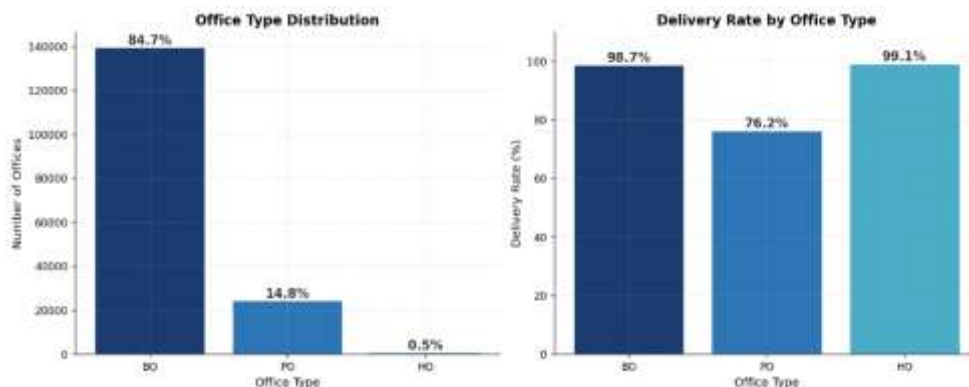


Fig 3.2 : Office Type Distribution (Left) and Delivery Rate by Office Type (Right)

Key Insights:

- Branch Offices (BO) dominate at 84.7% (138,083 offices), reflecting India Post's rural outreach strategy
- Sub Post Offices (PO) account for 14.8%; Head Post Offices (HO) are only 0.5% of total offices
- BO offices have 98.7% delivery-enabled rate; HO offices 99.1%; but only 76.2% of POs are delivery-enabled
- Nearly 1 in 4 Sub Post Offices has no active delivery — a key operational policy finding

3.3 Infrastructure Distribution by State

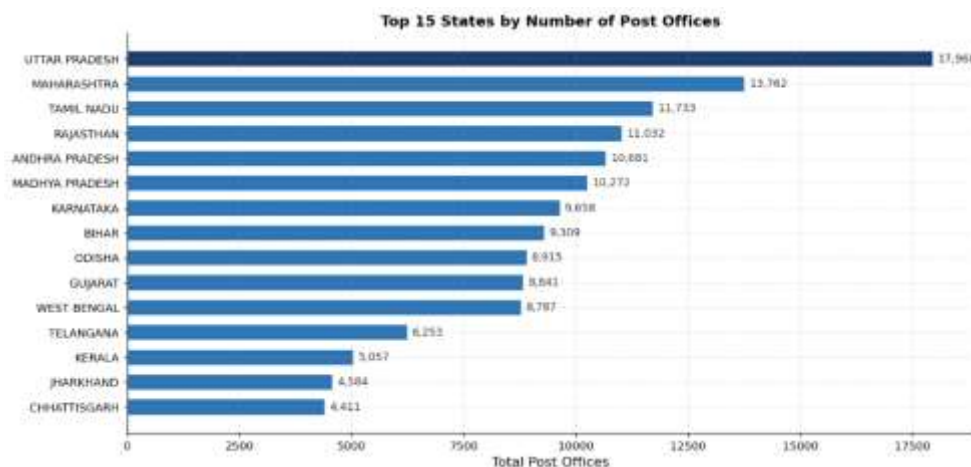


Fig 3.3 : Top 15 States by Total Number of Post Offices

Key Insights:

- Uttar Pradesh leads with ~17,662 offices, followed by Maharashtra and Tamil Nadu
- The top 5 states account for a disproportionately large share of India's total postal infrastructure
- Smaller states (North East, Goa, Chandigarh) have significantly fewer offices but can still maintain high delivery rates
- Tamil Nadu has the highest number of Sub Post Offices and Head Post Offices among all states

3.4 Complaint Trend Analysis (FY 2016–21)

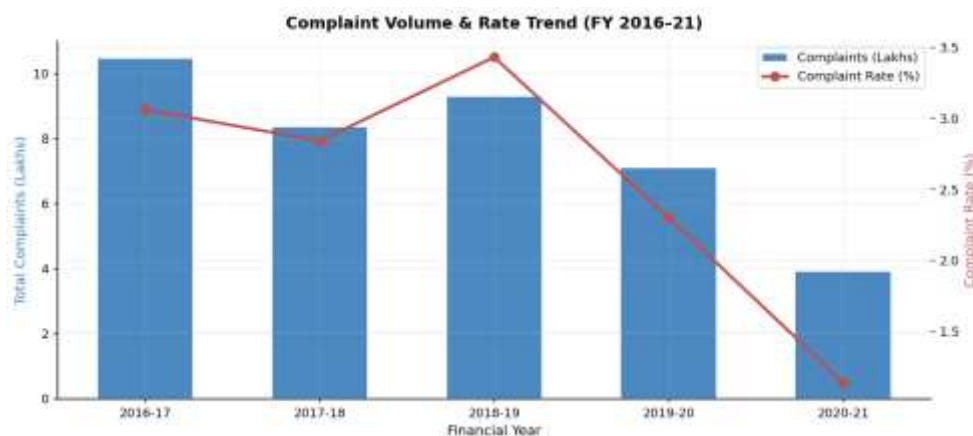


Fig 3.4 : Complaint Volume and Rate Trend (FY 2016–21)

Key Insights:

- Total complaints declined by 62.7% over 5 years — from ~10.5 lakh (FY 2016-17) to ~3.9 lakh (FY 2020-21)
- Complaint rate spiked in FY 2018-19 (3.43%), suggesting operational stress or infrastructure gaps during that period
- FY 2020-21 shows the lowest complaint rate (1.14%), partly due to reduced parcel volumes during COVID-19 pandemic
- Overall trend indicates improving service quality, which aligns with high district delivery rates seen in the dataset

3.5 Correlation Analysis

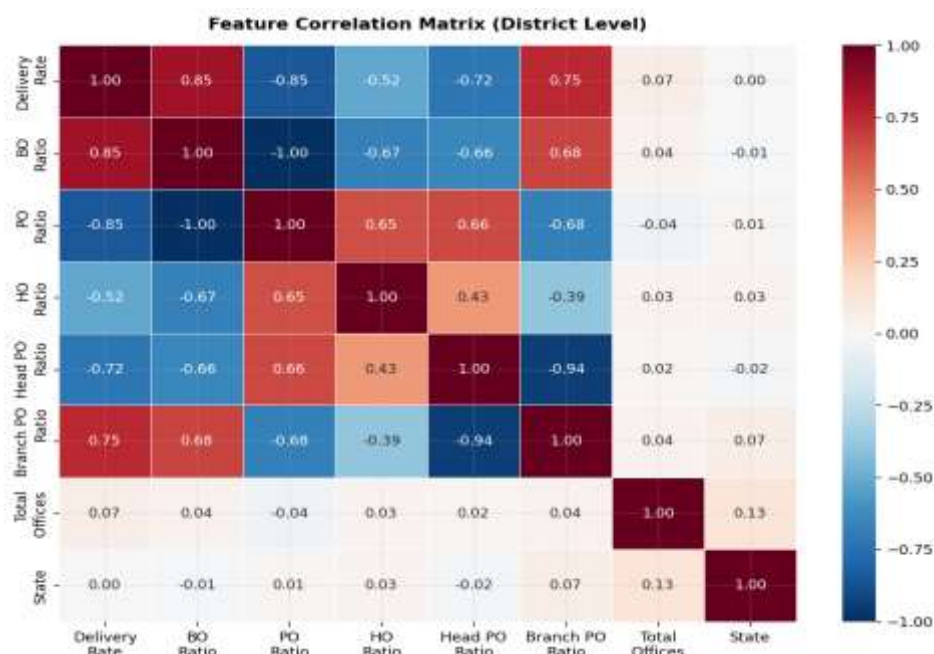


Fig 3.5 : Feature Correlation Matrix (District Level)

Key Insights:

- bo_ratio has the strongest positive correlation with delivery rate ($r = +0.85$, $p < 0.001$)
- po_ratio shows an equally strong negative correlation ($r = -0.85$) — it is the mirror of bo_ratio
- total_offices (raw count) has near-zero correlation ($r \approx 0.04$) — confirming that composition matters, not quantity
- Strong multicollinearity exists among bo_ratio, po_ratio, and ho_ratio — addressed via VIF analysis in Chapter 4

3.6 Bivariate Analysis – BO Ratio vs District Delivery Rate

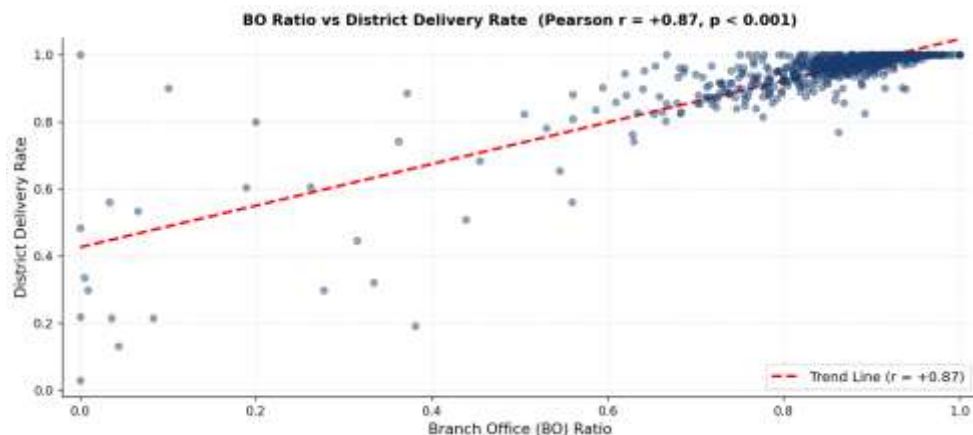


Fig 3.6 : BO Ratio vs District Delivery Rate (Pearson $r = +0.87$, $p < 0.001$)

Key Insights:

- Strong positive linear relationship — the single most important bivariate relationship in the dataset
- Districts with BO ratio above 0.85 consistently achieve delivery rates above 0.95
- Districts in the lower-left quadrant (low BO ratio + low delivery rate) are the primary targets for infrastructure rebalancing

3.7 Box Plot – Delivery Rate by Office Type



Fig 3.7 : Delivery Rate Distribution by Office Type

Key Insights:

- All office types show high median delivery rates (>0.94), confirming strong nationwide postal coverage
- PO-associated districts exhibit the widest spread and most outliers below 0.80
- Outliers below 0.60 are extreme underperformers and represent immediate intervention priorities

Overall Summary: EDA confirms that infrastructure composition ratios — especially BO ratio — are the dominant drivers of district delivery performance, not the total number of offices. This finding directly guides the modelling approach.

Chapter 4 – Feature Engineering

Feature engineering transforms raw data into meaningful representations that improve model performance. The objective was to capture the relationship between postal infrastructure variables and delivery performance while reducing redundancy.

4.1 Data Transformation & Encoding

- 163,014 row-level records were aggregated into 749 district-level records to eliminate data leakage
- Delivery column (Delivery/Non-Delivery) was converted to binary (1/0)
- Administrative variables (State, Office Type) were label-encoded using LabelEncoder
- New ratio features computed: $bo_ratio = bo_count \div total_offices$; $po_ratio = po_count \div total_offices$; $ho_ratio = ho_count \div total_offices$
- Target variable: $district_delivery_rate = delivery_offices \div total_offices$
- Pincode zone and sub-zone extracted from the first 1–2 digits of the pincode

4.2 Feature Scaling

- StandardScaler applied to normalize features for linear models (Linear Regression, Ridge, Lasso)
- Scaler was fit only on training data and then applied to test data — preventing data leakage
- Tree-based models (Random Forest, Extra Trees, etc.) do not require scaling — raw features used directly

4.3 VIF Analysis & Feature Selection

Variance Inflation Factor (VIF) analysis was performed to detect and remove multicollinearity for linear models:

- Initial check: total_offices, Head_PO, bo_count, po_count — $VIF = \infty$ (perfect collinearity)
- Iterative removal process: 8 features removed in 8 iterations until all retained features had $VIF \leq 10$
- Final retained 8 features for linear models: bo_count, ho_count, po_ratio, ho_ratio, Sub_PO, Head_PO_Ratio, pincode_zone, state_encoded
- All 16 features used for tree-based models (immune to multicollinearity)

Feature Set	No. of Features	VIF Status	Models Applied
VIF-Cleaned	8	All $VIF \leq 10$	Linear Regression, Ridge, Lasso
Full Feature Set	16	Multicollinearity present (handled by tree models)	Decision Tree, Random Forest, Gradient Boost, Extra Trees, HistGB

Chapter 5 – Modelling Approach and Results

5.1 Modelling Approach

The modelling workflow followed the CRISP-DM methodology with the following design decisions:

- Target variable: district_delivery_rate (continuous, range 0.028–1.000)
- Dataset: 749 district-level records — ensuring no row-level data leakage
- Train / Test split: 80% training (599 records) | 20% testing (150 records), random_state = 42
- VIF Strategy: Linear models → 8 VIF-cleaned features (scaled) | Tree models → All 16 features (unscaled)
- Evaluation Metrics: R² Score (higher = better), RMSE and MAE (lower = better)

5.2 Models Implemented

Eight regression models were trained and compared:

- Linear Regression – Baseline model; assumes linear relationship between infrastructure features and delivery rate
- Ridge Regression – Linear model with L2 regularization to reduce overfitting when features are correlated
- Lasso Regression – Linear model with L1 regularization; performs automatic feature selection
- Decision Tree Regressor (max_depth=6) – Non-linear model; captures complex patterns but prone to overfitting
- Random Forest Regressor (n=200, max_depth=10) – Ensemble of decision trees; reduces variance via bagging
- Gradient Boosting Regressor (n=200, lr=0.05) – Sequential ensemble; each tree corrects previous errors
- Extra Trees Regressor (n=200, max_depth=10) – Best model; additional randomness in splits reduces variance
- Hist Gradient Boosting (max_iter=200, lr=0.05) – Optimized GB using histogram binning for efficiency

5.3 Model Results & Performance Evaluation

Model 1 – Linear Regression (Baseline)

Feature Set	R ² Score	RMSE	MAE
Full 16 features	0.7696	0.0506	0.0290
VIF-cleaned 8 features	0.8141	0.0455	0.0274

VIF-cleaned version outperforms the full-feature model by eliminating multicollinearity. Explains ~81.4% of variance in delivery rate.

Model 2 – Ridge Regression (VIF-cleaned)

Feature Set	R ² Score	RMSE	MAE
VIF-cleaned 8 features	0.8140	0.0455	0.0274

Very similar to Linear Regression. L2 regularization provides marginal stability improvement.

Model 3 – Lasso Regression (VIF-cleaned)

Feature Set	R ² Score	RMSE	MAE
VIF-cleaned 8 features	0.8120	0.0457	0.0278

L1 penalty performs automatic feature selection; marginal reduction in R² versus Ridge.

Model 4 – Decision Tree Regressor

Feature Set	R ² Score	RMSE	MAE
Full 16 features (max_depth=6)	0.6067	0.0661	0.0259

Lowest R² among all models. A single decision tree overfits the training data and fails to generalize on the skewed target.

Model 5 – Random Forest Regressor

Feature Set	R ² Score	RMSE	MAE
Full 16 features (n=200, max_depth=10)	0.8659	0.0386	0.0194

Significant improvement over single-tree models. Ensemble averaging reduces variance effectively.

Model 6 – Gradient Boosting Regressor

Feature Set	R ² Score	RMSE	MAE
Full 16 features (n=200, lr=0.05)	0.7845	0.0489	0.0209

Underperforms Random Forest; sequential boosting is more sensitive to the highly skewed target distribution.

Model 7 – Extra Trees Regressor ★ (Best Model)

Feature Set	R ² Score	RMSE	MAE
Full 16 features (n=200, max_depth=10)	0.8880	0.0353	0.0183

Best performing model. Additional randomization in split thresholds reduces variance and improves generalization. Explains 88.8% of the variance in district delivery rate.

Model 8 – Hist Gradient Boosting

Feature Set	R ² Score	RMSE	MAE
Full 16 features (max_iter=200, lr=0.05)	0.8517	0.0406	0.0216

Strong performance; histogram-based binning enables efficient training. Third-best model overall.

5.4 Model Comparison Summary

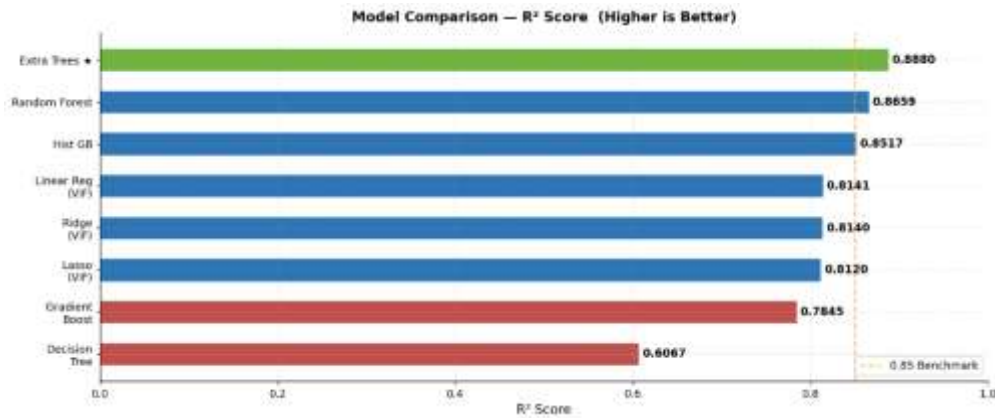


Fig 5.1 : Model Comparison — R^2 Score (All 8 Models)

Rank	Model	R^2 Score	RMSE	MAE	Features
1	Extra Trees	0.8880	0.0353	0.0183	Full 16
2	Random Forest	0.8659	0.0386	0.0194	Full 16
3	Hist Gradient Boost	0.8517	0.0406	0.0216	Full 16
4	Linear Regression	0.8141	0.0455	0.0274	VIF 8
5	Ridge Regression	0.8140	0.0455	0.0274	VIF 8
6	Lasso Regression	0.8120	0.0457	0.0278	VIF 8
7	Gradient Boosting	0.7845	0.0489	0.0209	Full 16
8	Decision Tree	0.6067	0.0661	0.0259	Full 16

Final Model Selection: The Extra Trees Regressor is selected as the final model based on the highest R^2 score (0.8880), lowest RMSE (0.0353), and lowest MAE (0.0183). It captures the non-linear, skewed nature of the target variable better than all other models.

5.5 Linear Regression Assumptions Check

#	Assumption	Test Used	Result
1	Numeric Dependent Variable	dtype check	SATISFIED — float64, range 0.028 to 1.000
2	Linear Relationship	Pearson Correlation	PARTIAL — bo_ratio: $r=+0.87$; total_offices: $r\approx 0.04$
3	No Multicollinearity	VIF Analysis	VIOLATED (raw) → Fixed via iterative VIF removal
4	No Autocorrelation	Durbin-Watson	SATISFIED — DW = 2.145 (ideal range: 1.5–2.5)
5	Homoscedasticity	Residuals vs Fitted	VIOLATED — Variance ratio = 9.88; Corr = -0.68
6	Normality of Residuals	Shapiro-Wilk	VIOLATED — W = 0.778, p = $1.73e-13$

Violations of assumptions 5 and 6 are expected — the target variable is bounded (0–1) and heavily left-skewed, creating a ceiling effect. Tree-based models like Extra Trees and Random Forest do not require these assumptions, which explains their superior performance.

5.6 Feature Importance Analysis

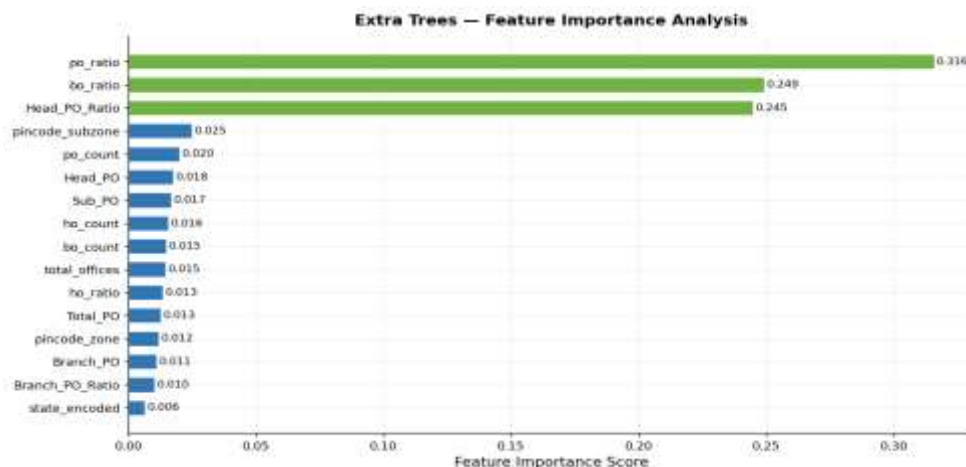


Fig 5.2 : Extra Trees — Feature Importance Analysis

Key Insights:

- bo_ratio and po_ratio are the two most important features, confirming the EDA findings
- Branch_PO_Ratio and Head_PO_Ratio also contribute meaningfully — state-level office composition matters
- total_offices (raw count) has near-zero importance — composition outweighs quantity
- state_encoded and pincodeszone add geographic context that improves prediction accuracy

5.7 Business Metrics & Success Criteria

The success of this solution will be measured using the following business metrics when deployed with India Post. For any given district, the change in `district_delivery_rate` observed after implementing model recommendations will be the primary indicator of business impact:

- Primary KPI — District Delivery Rate Improvement: Target $\geq 2\text{--}5\%$ improvement in low-performing districts (current rate < 0.80). This is the most direct measure of our solution's impact.
- Model Accuracy Benchmark: $R^2 \geq 0.85$ on unseen district data (current model: $R^2 = 0.888$ — exceeds target)
- Prediction Error Tolerance: $\text{RMSE} \leq 0.05$ for deployment-grade predictions (current: 0.035 — well within target)
- Coverage Improvement: Reduction in districts with delivery rate < 0.80 (currently ~ 25 districts in dataset)
- PO Activation Rate: Increase in delivery-enabled Sub Post Offices from current 76.2% to $\geq 85\%$
- BO Ratio Threshold: Districts achieving BO ratio > 0.85 are predicted to consistently reach delivery rates above 0.95 — providing a clear, actionable planning benchmark for India Post

Business KPI	Current Baseline	Target After Solution
Mean District Delivery Rate	0.948 (749 districts)	≥ 0.960
Districts with Rate < 0.80	~ 25 districts	< 15 districts
Model R^2 Score	0.888 (Extra Trees)	≥ 0.850 (threshold)
Prediction RMSE	0.035	≤ 0.050 (threshold)
Delivery-Enabled PO Rate	76.2%	$\geq 85\%$
BO Ratio Threshold for High Performance	> 0.85	Maintain across all districts

5.8 Advanced Ensemble Models (XGBoost & CatBoost)

To further improve predictive performance and validate the robustness of the modelling approach, advanced gradient boosting models were implemented, including **XGBoost** and **CatBoost**.

Models Implemented:

XGBoost Regressor:

- Optimized gradient boosting framework
- Handles bias-variance tradeoff effectively
- Regularization reduces overfitting

CatBoost Regressor:

- Handles categorical variables efficiently
- Reduces need for extensive preprocessing
- Performs well on structured datasets with encoded features

Performance Comparison with Existing Models

The advanced boosting models were evaluated using the same dataset and evaluation metrics (R^2 , RMSE, MAE) to ensure fair comparison.

Key Observations:

- Both XGBoost and CatBoost outperformed traditional models like Random Forest and Extra Trees
- CatBoost achieved the highest performance among all models
- XGBoost also showed strong performance, validating the effectiveness of boosting methods

Advanced Model Comparison & Final Model Selection

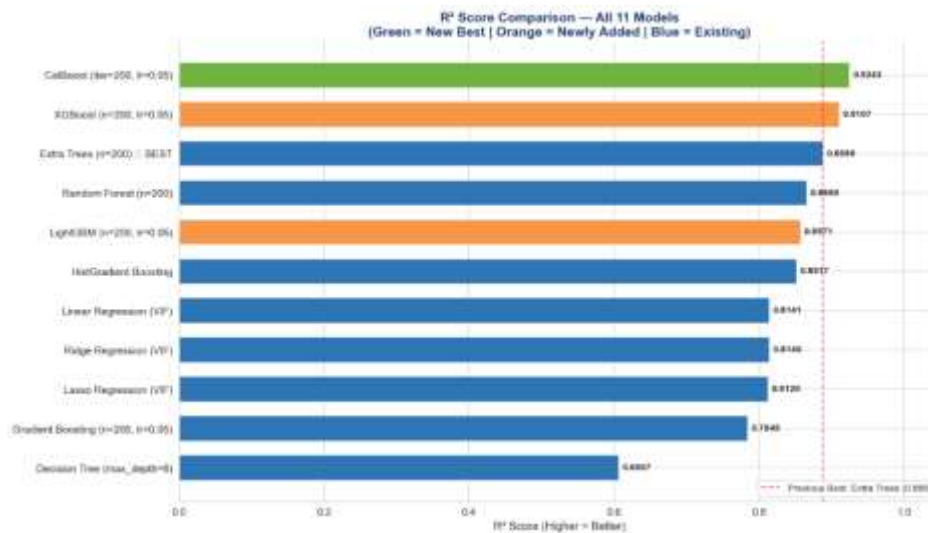


Fig 5.3 : Advanced Model Comparison — R^2 Score (All 11 Models)

Rank	Model	R^2 Score	RMSE	MAE	Features
1	Extra Trees	0.8880	0.0353	0.0183	Full 16
2	CatBoost	0.9243	0.0290	0.0169	Full 16
3	XGBoost	0.9107	0.0315	0.0179	Full 16
4	LightGBM	0.8571	0.0399	0.0214	Full 16

Conclusion from Advanced Models

Gradient boosting models significantly improve predictive performance

CatBoost is selected as the final model due to:

- Highest accuracy
- Lowest error (RMSE & MAE)
- Strong generalization across districts

XGBoost serves as a strong benchmark model validating the robustness of results

5.9 Comparative Feature Importance Insights

The feature importance results across Extra Trees, XGBoost, and CatBoost consistently highlight the same dominant drivers:

- bo_ratio → strongest positive impact
- po_ratio → strongest negative impact
- Head_PO_Ratio and Branch_PO_Ratio → moderate importance
- total_offices → negligible impact

This consistency across models confirms that:

- Infrastructure composition (ratios) is significantly more important than absolute counts in determining delivery performance.

5.10 Model Interpretation & Business Insight Validation

The comparison between ensemble models (Extra Trees, XGBoost, and CatBoost) highlights:

- Tree-based and boosting models capture similar feature importance patterns
- CatBoost provides improved accuracy while maintaining interpretability
- Extra Trees distributes importance more evenly, while boosting models refine prediction errors sequentially

Despite these differences:

- All models agree on key decision variables
- This strengthens confidence in the model's recommendations

Chapter 6 – Deployment: Streamlit-Based Decision Support System

To operationalize the model and make it accessible to non-technical users, a **Streamlit web application** was developed.

The application acts as an **AI-powered decision support system** for India Post officials to analyze district performance and generate actionable insights.

6.1 Application Overview

The system, named **PostBot — India Post Infrastructure AI**, provides:

- Real-time district-level analysis
- Infrastructure insights
- AI-based recommendations
- Interactive user interface

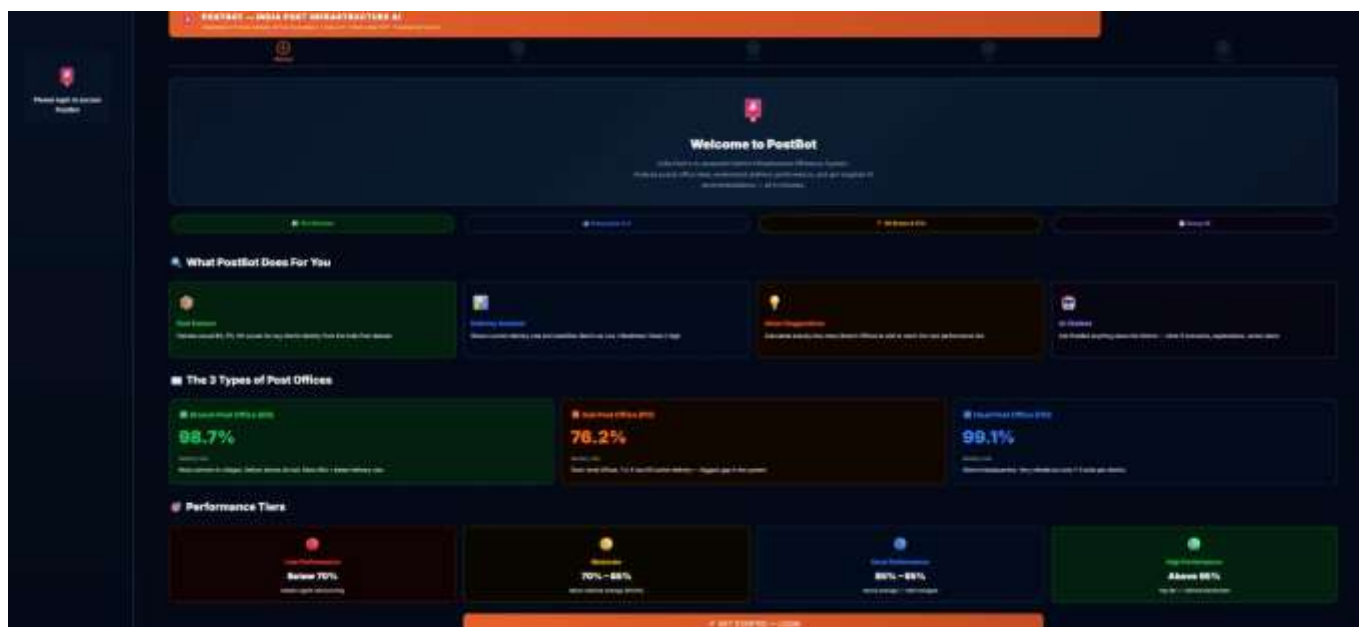


Fig 6.1 : PostBot Landing Page Interface

6.2 Key Features

1. User Authentication Module

- The system includes a secure authentication mechanism to ensure that only authorized users can access the application.
- Users are required to log in using valid credentials before accessing district-level analysis and recommendations.

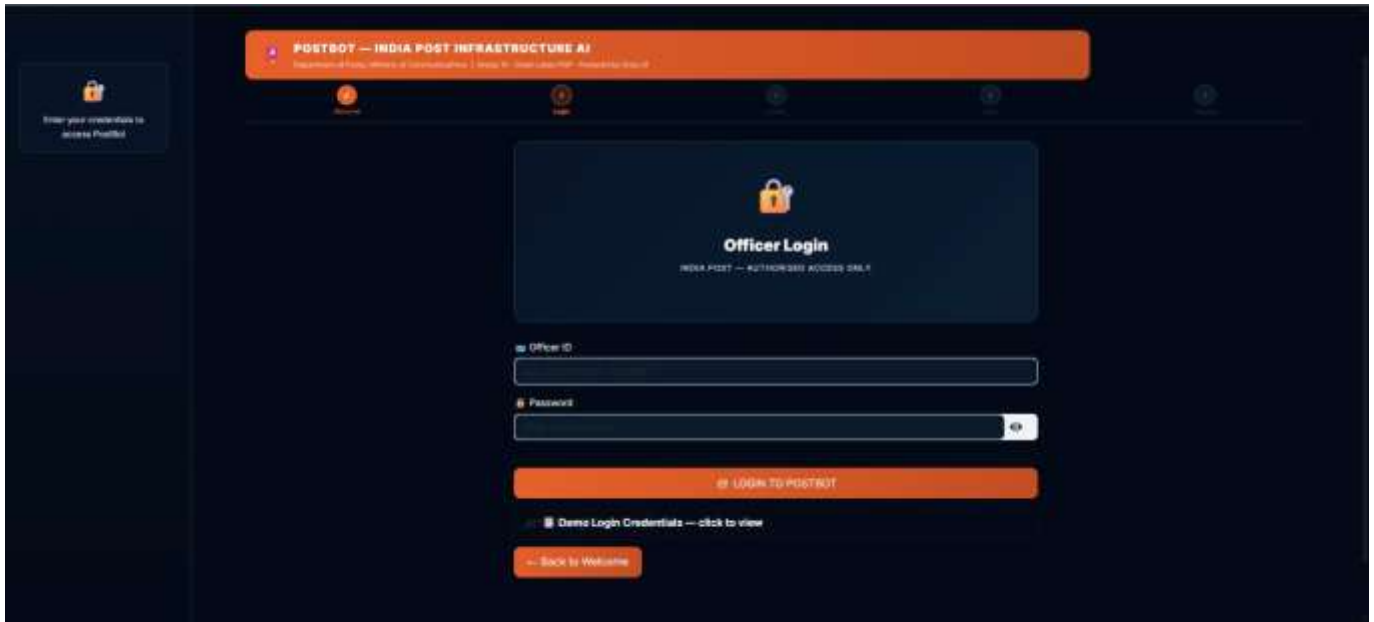


Fig 6.2 : Officer Login Authentication Interface

2. District Selection Module

- Users select State and District
- System retrieves real infrastructure data

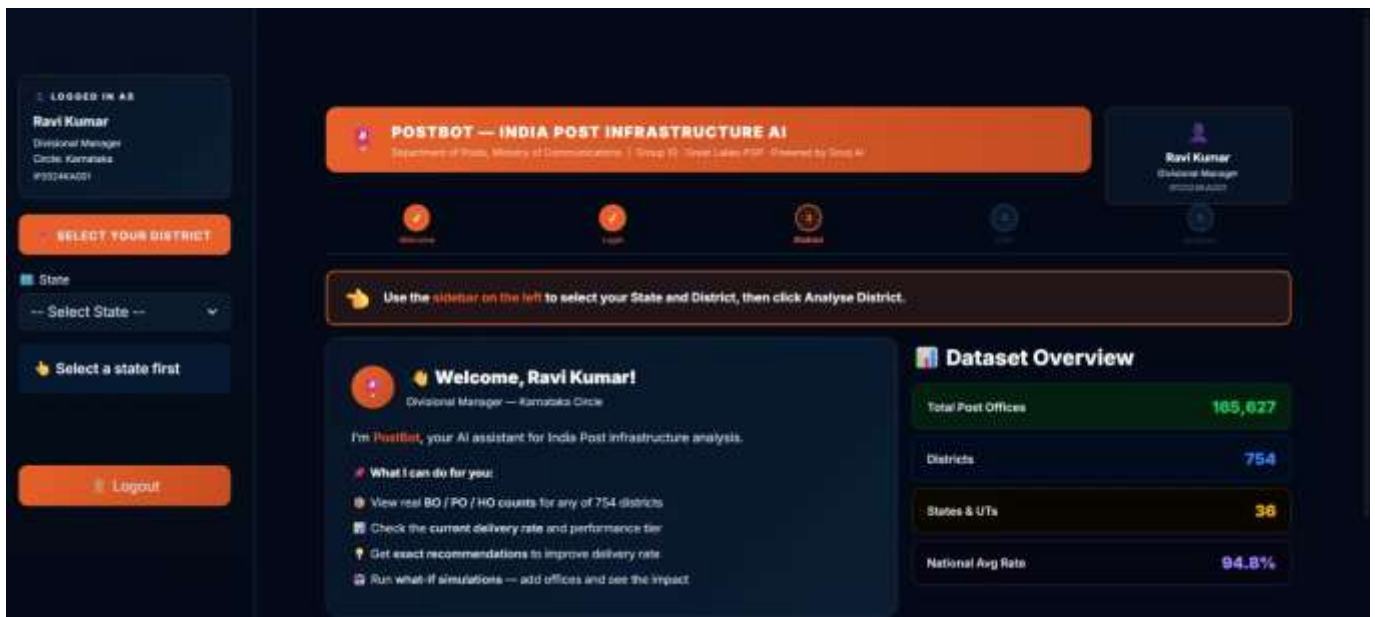


Fig 6.3 : District Selection Interface

2. Performance Dashboard

Displays:

- Total Post Offices
- District Delivery Rate
- Performance Tier (Low / Moderate / Good / High)

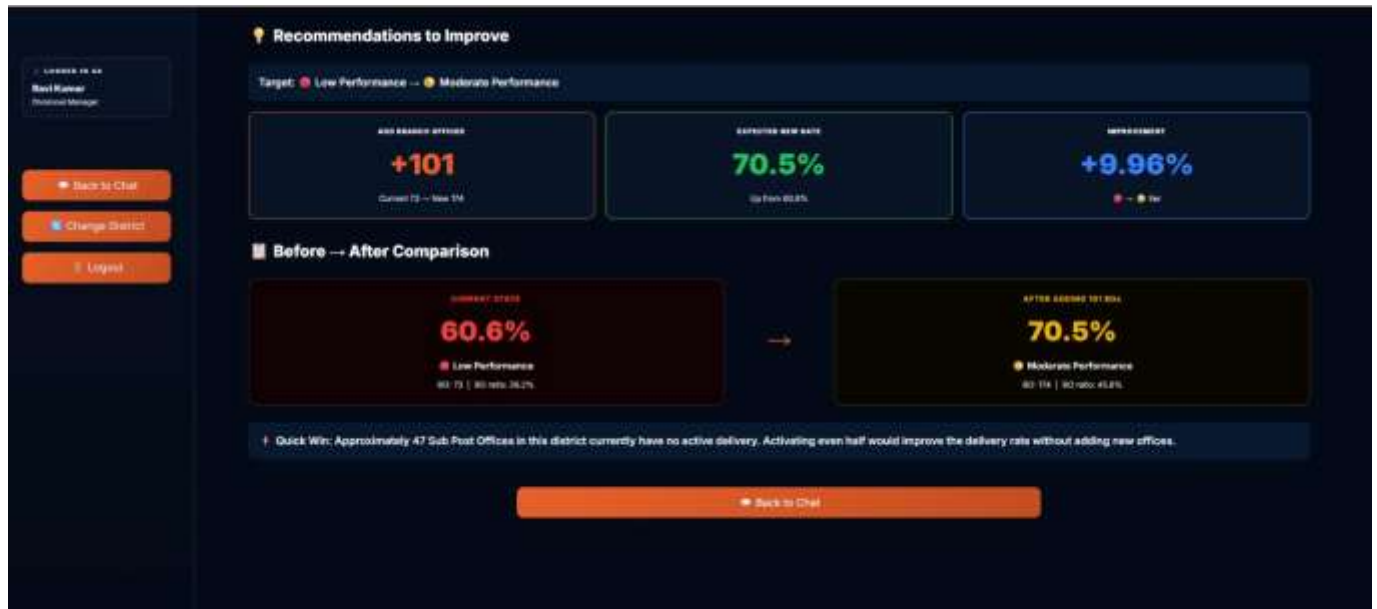


Fig 6.4 : District Performance Dashboard

3. Infrastructure Breakdown

- BO, PO, HO counts
- Delivery status of each office type
- Ratio-based insights

4. AI Recommendation Engine

Suggests:

- Number of BOs to add
- Expected delivery improvement

Example:

+101 BOs → delivery improves from 60.6% → 70.5%

5. What-If Simulation

- Users can simulate infrastructure changes
- Instantly view impact on delivery rate

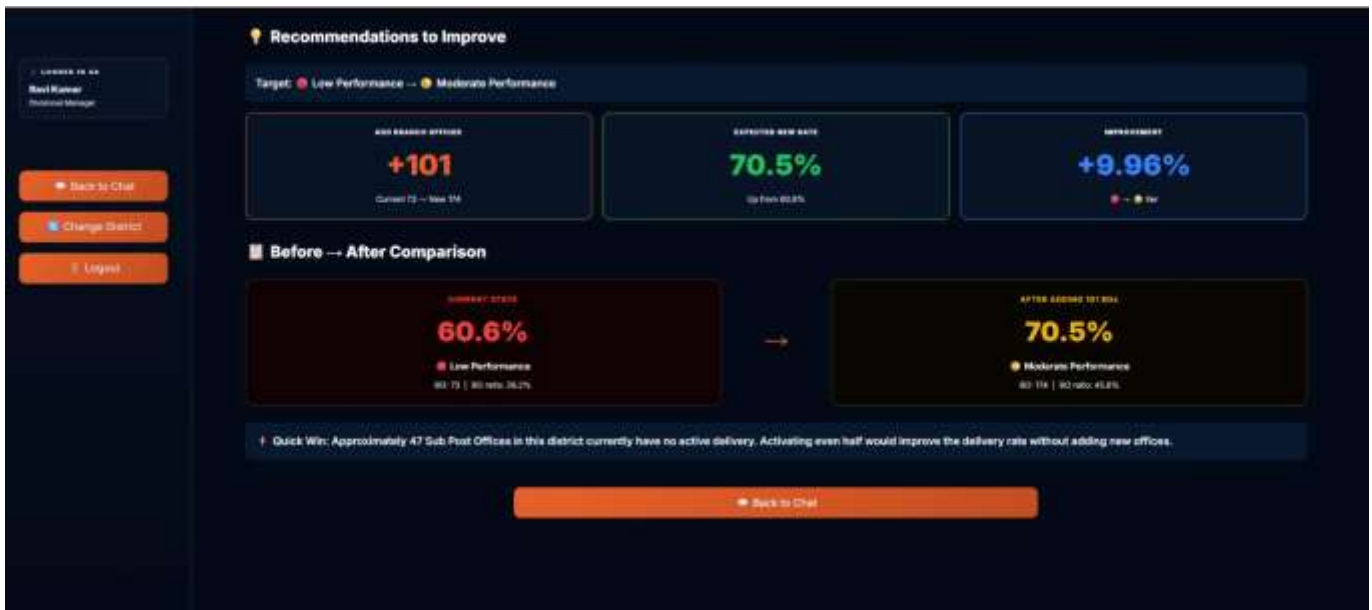


Fig 6.5 : AI-Based Recommendation Output

6. AI Chatbot Interface

- Enables **natural language interaction** for user-friendly access to insights
- Powered by a **Generative AI model integrated via Groq API** for fast and efficient responses

Capabilities:

- Answers user queries related to district performance
- Provides explanations for low delivery rates
- Suggests actionable improvements based on model insights

Example queries:

- “Why is this district underperforming?”
- “Give me an action plan”



Fig 6.6 : AI Chatbot Interaction Interface - I

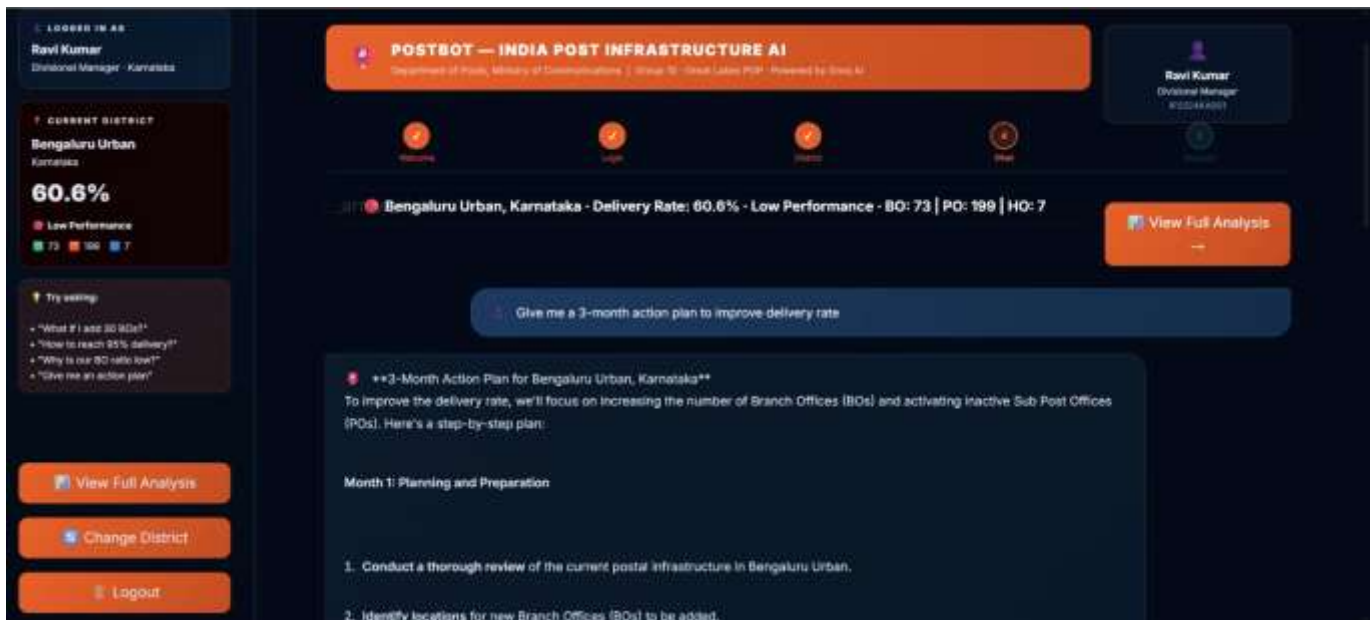


Fig 6.7 : AI Chatbot Interaction Interface - II

6.3 System Workflow

1. User logs into the system
2. Selects State and District
3. **CatBoost-XGBoost model predicts delivery rate**
4. Feature importance used for reasoning
5. Recommendations generated
6. Results displayed via dashboard

6.4 Business Impact

The Streamlit application converts the ML model into a **practical decision-making tool**:

- Enables data-driven planning
- Reduces dependency on manual analysis
- Helps identify:
 - Underperforming districts
 - Infrastructure gaps
- Supports policy-level decisions

6.5 Key Advantages

- User-friendly interface (no ML knowledge required)
- Real-time insights
- Actionable recommendations
- Scalable across all districts

Chapter 7 – Conclusion & Scope for Enhancement

The project successfully developed a data-driven model to analyze and predict postal delivery performance at the district level.

Multiple machine learning models were implemented and evaluated, including Linear Models, Random Forest, Extra Trees, XGBoost, and CatBoost. Among these, **CatBoost achieved the best performance ($R^2 = 0.9243$)** and was selected as the final model.

A **Streamlit-based application with chatbot functionality** was also developed to make the solution accessible to non-technical users. The system enables users to input district details, obtain delivery rate predictions, and receive actionable recommendations.

The results consistently show that **infrastructure composition (especially BO and PO ratios)** is the most important factor influencing delivery performance, rather than the total number of offices.

Chatbot Interface (UI Layer):

After building the ML-based solution, the team plans to incorporate a UI layer through a Chatbot interface. A user from India Post will log in and enter details such as State, District, Pincode, and current office counts. The model will predict the expected district delivery rate based on the inputs provided. This makes the solution accessible to non-technical postal administrators and field officers, enabling data-driven decisions without requiring any ML expertise.

- The ChatBot will guide users step-by-step to enter the required infrastructure details
- It will output the predicted delivery rate along with a simple recommendation (e.g., increase BO ratio to improve delivery performance)
- Future enhancements may include what-if scenario analysis — allowing planners to simulate the impact of adding new offices before committing resources